Week 11 Progress Report

Details

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- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week:** **(16/03/2026 to 20/03/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of the Week

Week 11 continued the hands-on lab work from Week 10. Two main tasks were completed this week: permanently fixing the dynamic IP problem on the Ubuntu Server VM, and running the second attack simulation in the lab using Wazuh's File Integrity Monitoring (FIM) module.

In Week 10, the Ubuntu Server VM was getting a new IP address every time it was rebooted. This was causing the Windows Wazuh agent to lose connection to the SIEM manager. This week, a static IP address (192.168.1.6) was configured on the Ubuntu Server using Netplan by creating a new configuration file called 99-static-ip.yaml. After applying the configuration and rebooting the VM, the IP remained fixed at 192.168.1.6. This issue is now permanently resolved and no manual configuration will be needed in future sessions.

With the server IP stable, the second attack simulation was performed: a File Integrity Monitoring (FIM) trigger. A new test file named test-wazuh.txt was created inside the C:\Windows\System32\drivers\etc directory, which is actively monitored by Wazuh. Wazuh detected the file creation in real-time, generated Rule 554 at Level 5 severity, and captured the file hash in the alert metadata. The alert was verified in the Wazuh Security Events dashboard, confirming that Wazuh's FIM module is correctly monitoring filesystem changes on the Windows endpoint.

Objectives of the Week

- To permanently resolve the dynamic IP issue on the Ubuntu Server VM by configuring a static IP address (192.168.1.6) using Netplan.

- To verify that the static IP persists after a reboot and that the Windows Wazuh agent remains connected without any manual changes.
- To perform Attack Simulation 2: File Integrity Monitoring (FIM) trigger — create a test file inside a Wazuh-monitored directory on the Windows endpoint and verify real-time detection.
- To confirm that Wazuh generates the correct FIM alert (Rule 554, Level 5) and captures the file hash in the alert metadata.
- To investigate the generated alert in the Wazuh Security Events dashboard and confirm the detection details.

Key Accomplishments

- Configured a static IP address (192.168.1.6) on the Ubuntu Server VM using Netplan by creating a new configuration file: /etc/netplan/99-static-ip.yaml.
- Applied the configuration using 'sudo netplan apply', rebooted the VM, and confirmed the static IP persisted using 'ip a'.
- The Windows Wazuh agent automatically reconnected to the SIEM manager at the fixed IP after reboot — no manual ossec.conf update required.
- **Executed Attack Simulation 2: File Integrity Monitoring (FIM) Trigger** — created test-wazuh.txt inside C:\Windows\System32\drivers\etc on the Windows 11 endpoint.
- Wazuh detected the file creation in real-time and triggered Rule 554 at severity Level 5, confirming the FIM module is actively monitoring the filesystem.
- Alert metadata captured in the Wazuh Security Events dashboard included: file path, event type (file added), file hash (MD5/SHA1), and timestamp.
- Full alert investigation completed in the Wazuh dashboard — verified rule description, agent details, and file integrity event details.

Issues Faced & Resolutions – Week 11

1. Ubuntu Server dynamic IP changing after every reboot

Issue: The Ubuntu Server VM was receiving a new dynamic IP address every time it was rebooted. This was causing the Windows Wazuh agent to lose connection to the SIEM manager after each restart, requiring manual updates to the agent configuration file (ossec.conf) every session.

Resolution: Created a new Netplan configuration file at /etc/netplan/99-static-ip.yaml and defined a static IP address of 192.168.1.6 for the enp0s3 interface. Applied the configuration using 'sudo netplan apply'. Rebooted the VM and verified the IP address remained 192.168.1.6 using 'ip a'. The Windows agent reconnected automatically. This issue is permanently resolved.

2. Confirming FIM is monitoring the correct directory

Issue: Before running the FIM simulation, it was necessary to confirm that the C:\Windows\System32\drivers\etc directory was already included in Wazuh's active FIM monitoring configuration, so that a file creation event would actually trigger an alert.

Resolution: Verified the Wazuh agent's ossec.conf on the Windows endpoint to confirm the directories section included system paths under real-time monitoring. Created test-wazuh.txt in the target directory and confirmed the alert appeared in the Wazuh dashboard within seconds, validating the FIM configuration.

Tools Used – Week 11

- **Wazuh SIEM (Manager + Indexer + Dashboard v4.7.5)** – Primary detection and monitoring platform; used to detect FIM events, generate alerts, and investigate rule triggers.
- **Netplan (Ubuntu Server 24.04)** – Network configuration tool used to assign a static IP address (192.168.1.6) to the Ubuntu Server VM by creating 99-static-ip.yaml.
- **Ubuntu Server 24.04 (VM)** – Hosts the Wazuh SIEM stack; static IP configured this week to permanently fix the reboot IP-change issue.
- **VirtualBox** – Hypervisor hosting the Ubuntu Server 24.04 VM and Ubuntu 22.04 attacker VM on an internal network.
- **Wazuh Agent 4.7.5 (Windows 11 Endpoint)** – Installed on the host machine; monitors filesystem changes via the FIM module and forwards events to the SIEM.
- **Windows File System (C:\Windows\System32\drivers\etc)** – Target directory for the FIM simulation; test-wazuh.txt was created here to trigger the Wazuh FIM alert.

Conclusion of Week 11

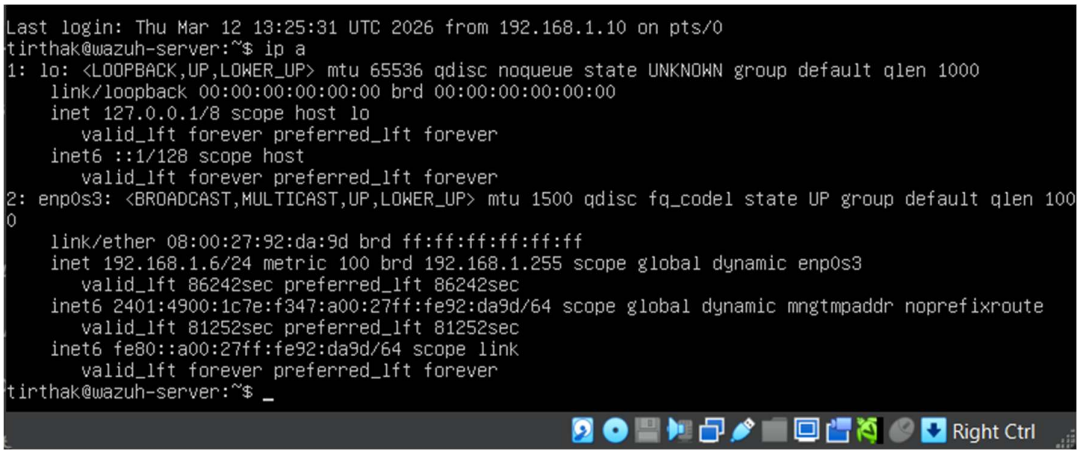
Week 11 delivered two important results for the SOC Analyst home lab project. First, the static IP configuration using Netplan permanently resolved the recurring dynamic IP problem that had been disrupting the lab setup since Week 10. The Ubuntu Server VM now boots with a fixed IP (192.168.1.6) every time, and the Windows Wazuh agent connects automatically without any manual intervention. This makes the lab stable and ready for all future simulations.

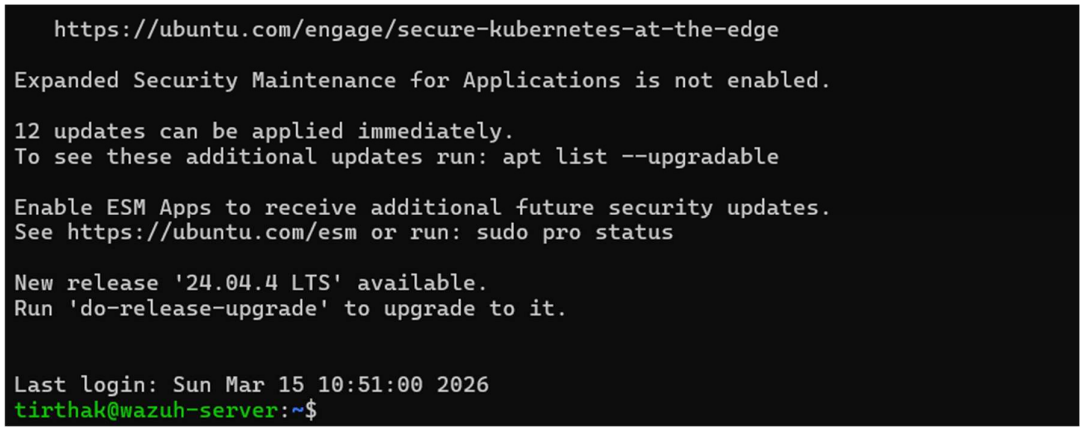
Second, Attack Simulation 2 — the File Integrity Monitoring (FIM) trigger — was successfully executed and detected by Wazuh in real-time. The creation of test-wazuh.txt inside the monitored C:\Windows\System32\drivers\etc directory immediately triggered Rule 554 at Level 5, with the file hash captured in the alert metadata. This demonstrates that Wazuh is not only detecting authentication-based attacks (as shown in Week 10 with Event ID 4625) but also monitoring filesystem changes, which is a critical detection capability for insider threats and malware activity.

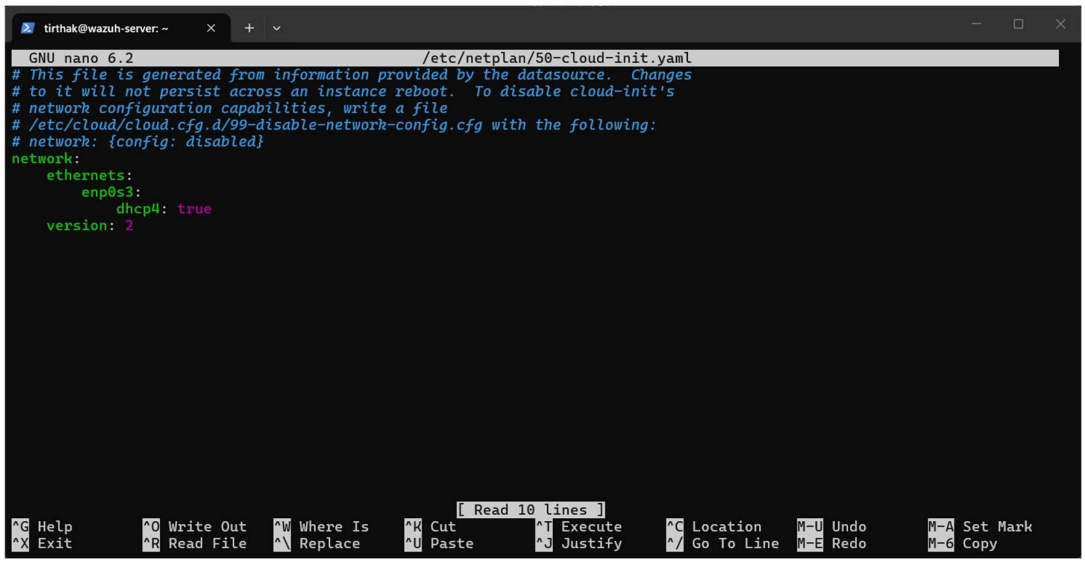
Together, the Week 10 and Week 11 simulations — Failed Windows Login (T1078, Rule 60122) and FIM File Creation (Rule 554) — form the core detection demonstrations for the CA2 mid-term presentation. The lab is now well-positioned for the next phase of work in Week 12, which will include the SSH brute-force simulation using Hydra from the attacker VM.

Screenshots of Activities (if Any)

[Please insert screenshots from your Wazuh dashboard here — Static IP via netplan, ip a output showing 192.168.1.6, Windows agent active, test-wazuh.txt creation, Rule 554 Level 5 FIM alert, file hash in alert metadata]







```yaml
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: false
      addresses:
        - 192.168.[redacted]
      routes:
        - to: default
          via: 192.168.[redacted]
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

```
tirthak@wazuh-server: ~
tirthak@wazuh-server:~$ sudo nano /etc/netplan/99-static-ip.yaml
tirthak@wazuh-server:~$ sudo netplan apply

** (generate:3450): WARNING **: 10:59:47.941: Permissions for /etc/netplan/99-static-ip.yaml are too open. Netplan confi
guration should NOT be accessible by others.
WARNING:root:Cannot call Open vSwitch: ovsdb-server.service is not running.

** (process:3448): WARNING **: 10:59:48.743: Permissions for /etc/netplan/99-static-ip.yaml are too open. Netplan config
uration should NOT be accessible by others.

** (process:3448): WARNING **: 10:59:49.049: Permissions for /etc/netplan/99-static-ip.yaml are too open. Netplan config
uration should NOT be accessible by others.

** (process:3448): WARNING **: 10:59:49.049: Permissions for /etc/netplan/99-static-ip.yaml are too open. Netplan config
uration should NOT be accessible by others.
tirthak@wazuh-server:~$
```

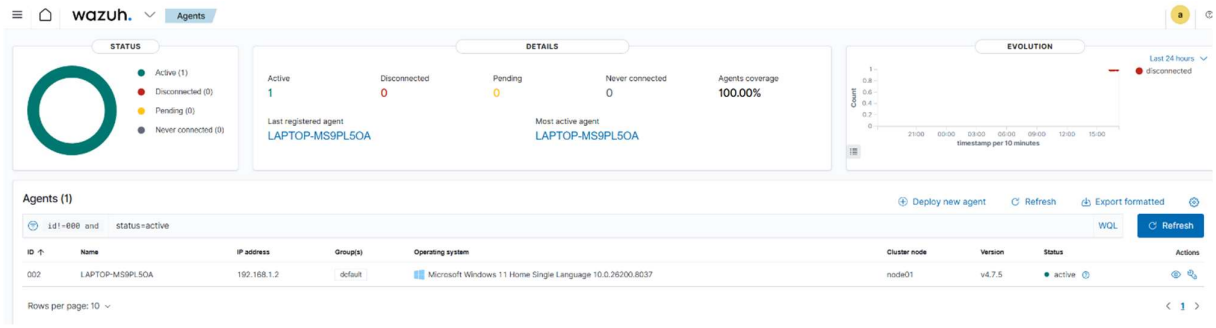
```
tirthak@wazuh-server: ~
tirthak@wazuh-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:92:da:9d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.6/24 brd 192.168.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 2401:4900:1c7e:f347:a00:27ff:fe92:da9d/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 80592sec preferred_lft 80592sec
    inet6 fe80::a00:27ff:fe92:da9d/64 scope link
       valid_lft forever preferred_lft forever
tirthak@wazuh-server:~$
```

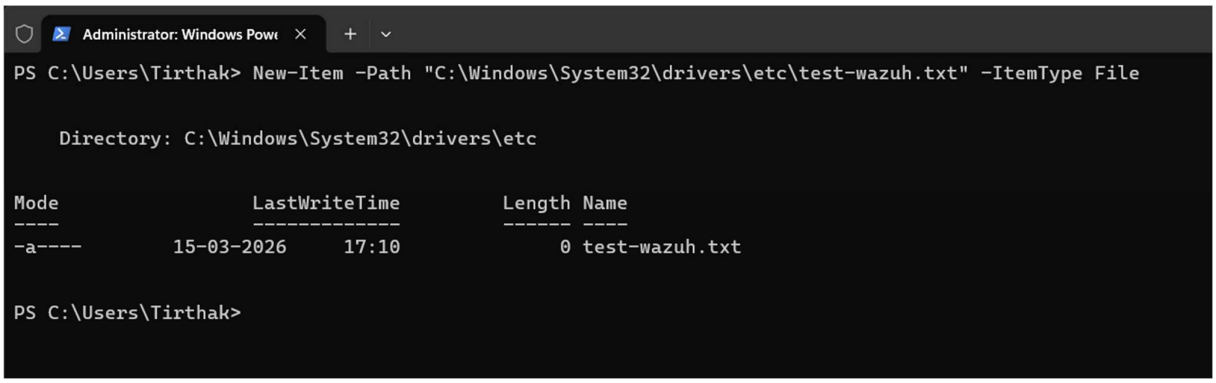
```
tirthak@wazuh-server: ~
tirthak@wazuh-server:~$ sudo systemctl is-active wazuh-manager
sudo systemctl is-active wazuh-indexer
sudo systemctl is-active wazuh-dashboard
[sudo] password for tirthak:
active
active
active
tirthak@wazuh-server:~$ Start-Service wazuhsvc
Get-Service wazuhsvc
Start-Service: command not found
Get-Service: command not found
tirthak@wazuh-server:~$
```

```
PS C:\Users\Tirthak> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\Users\Tirthak> Restart-Service wazuhsvc
PS C:\Users\Tirthak>
```

Alert Summary — What happened:

Field	Value	What it means
Event ID	4625	Windows failed logon
Target User	Tirthak	Your account was targeted
Rule ID	60122	Logon failure rule fired
Rule Level	5	Medium severity
Fired Times	6	6 failed attempts
Logon Type	2	Interactive (someone at keyboard)
Failure Reason	Unknown user or bad password	Wrong password entered
MITRE Technique	T1078 — Valid Accounts	Attacker trying known usernames
MITRE Tactic	Initial Access, Privilege Escalation	Real attack classification
Compliance	GDPR, HIPAA, PCI-DSS, NIST 800-53	Regulatory frameworks triggered





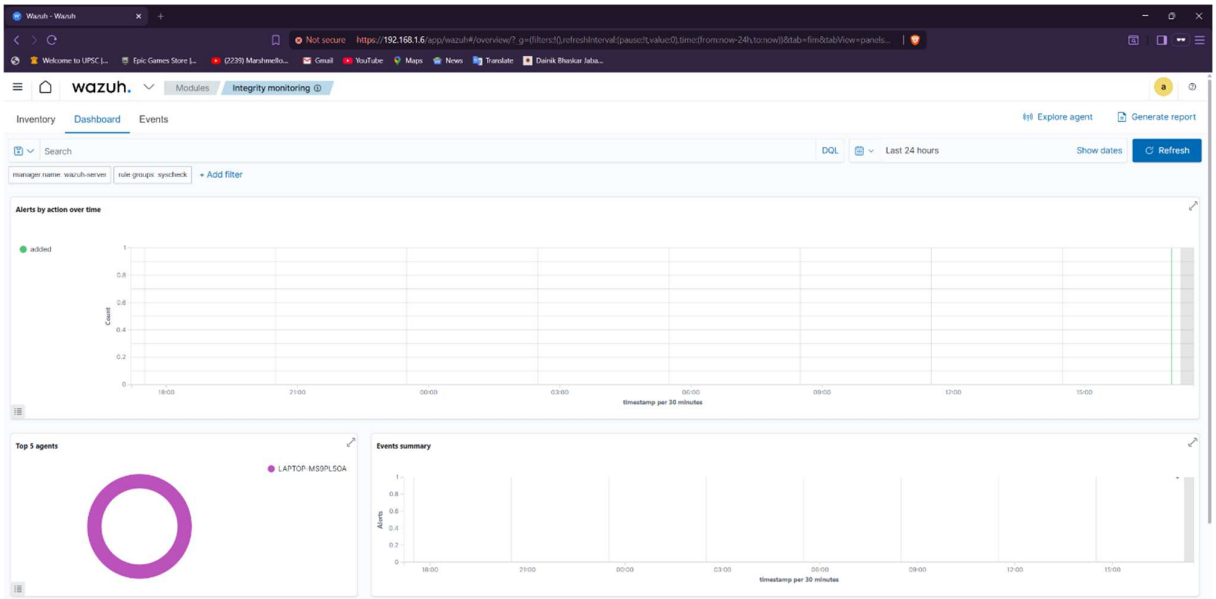


Fig: File integration Monitoring -test file simulation had been performed but its taking time to show in the SIEM dashboard logs.

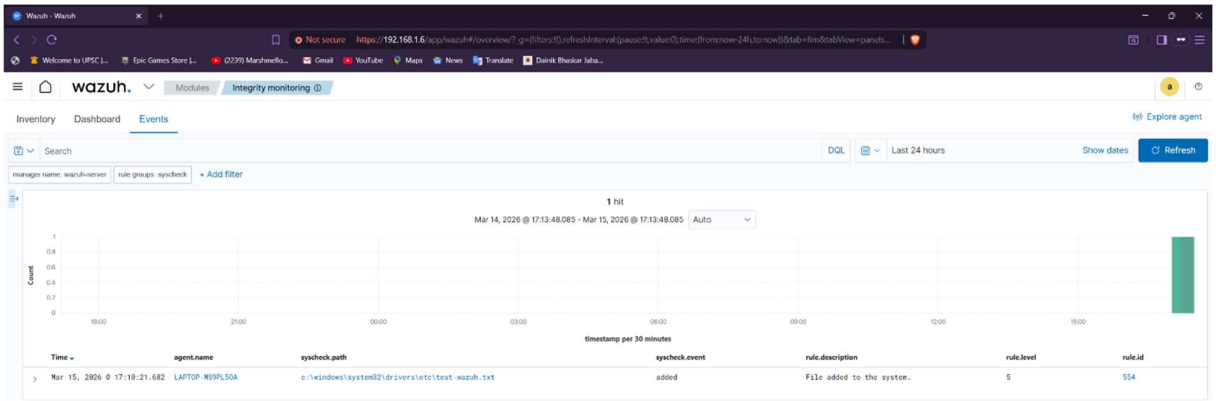


Fig : As you can see we got one hit for the test file that was found in confidential configuration files which should not be configured as it trigures the FIM attack simulation protocol

What Wazuh detected: A new file was created at `C:\Windows\System32\drivers\etc\test-wazuh.txt` and Wazuh caught it in **realtime** — not on a scheduled scan.

Key Alert Details:

Field	Value	What it means
Rule ID	554	Wazuh's rule for "file added"
Rule Level	5	Medium severity
Event	added	New file was created
Mode	**realtime**	Detected instantly, not on schedule
File Owner	Administrators	Created by admin account
File Size	0 bytes	Empty file
MD5 Hash	d41d8cd98f00b204e9800998ecf8427e	File fingerprint (empty file hash)

Compliance frameworks triggered:

Framework	Tag	Meaning
GDPR	II_5.1.f	Data integrity monitoring
HIPAA	164.312.c.1/c.2	Integrity controls
PCI-DSS	11.5	File change detection
NIST 800-53	SI.7	System integrity

Why this matters for SOC: In a real attack, an attacker might drop a malicious file into a sensitive Windows directory like `System32`. FIM would catch it immediately — just like it caught your test file. The file hash (MD5/SHA1/SHA256) is especially important — SOC analysts compare these against threat intelligence databases to check if a file is known malware.